

Ex.No.1
09.01.2025

Data Definition Language Commands and Integrity Constraints

AIM:

To execute Data Definition Languages Commands and Integrity Constraints

1.CREATE TABLE:

```
SQL> create table stu_details(rollno number(4) primary key,name varchar2(20) not null,dept varchar2(20) not null,year number(1) not null);
```

Table created.

SCHEMA OF THE TABLE:

```
SQL> desc stu_details;
```

Name	Null?	Type
-----	-----	-----
ROLLNO	NOT NULL	NUMBER(4)
NAME	NOT NULL	VARCHAR2(20)
DEPT	NOT NULL	VARCHAR2(5)
YEAR	NOT NULL	NUMBER(1)

INSERT VALUES INTO TABLE:

(i) THROUGH SYSTEM:

```
SQL> insert into stu_details values(&rollno, '&name', '&dept', &year);
```

Enter value for rollno: 180

Enter value for name: Yokarajan

Enter value for dept: IT

Enter value for year: 2

```
old 1: insert into stu_details values(&rollno, '&name', '&dept', &year)
```

```
new 1: insert into stu_details values(180, 'Yokarajan', 'IT', 2)
```

1 row created.

Enter value for rollno: 170

Enter value for name: viji

Enter value for dept: cse

Enter value for year: 3

```
old 1: insert into stu_details values(&rollno, '&name', '&dept', &year)
new 1: insert into stu_details values(170, 'viji', 'cse', 3)
1 row created.
```

Enter value for rollno: 137

Enter value for name: Ros

Enter value for dept: AIDS

Enter value for year: 1

```
old 1: insert into stu_details values(&rollno, '&name', '&dept', &year)
new 1: insert into stu_details values(137, 'Ros', 'AIDS', 1)
1 row created.
```

(ii)MANUAL INPUT:

```
SQL> insert into stu_details values(131, 'rithi', 'MTS', 2);
1 row created.
SQL> insert into stu_details values(134, 'gojo', 'FT', 3);
1 row created.
```

```
select * from stu_details;
```

ROLLNO	NAME	DEPT	YEAR
180	Yokarajan	IT	2
170	viji	CSE	3
137	Ros	AIDS	1
131	rithi	MTS	2
134	gojo	FT	3

2.ALTER:

ADD ATTRIBUTE

BEFORE ADDING ATTRIBUTE

```
SQL> desc stu_details;
```

Name	Null?	Type
ROLLNO	NOT NULL	NUMBER(4)

NAME	NOT NULL	VARCHAR2(20)
DEPT	NOT NULL	VARCHAR2(5)
YEAR	NOT NULL	NUMBER(1)

AFTER ADDING ATTRIBUTE

SQL> alter table stu_details add(hostel_default number(3) default yes);
Table altered.

AFTER ADDING ATTRIBUTE:

SQL> desc stu_details;

Name	Null?	Type
-----	-----	-----
ROLLNO	NOT NULL	NUMBER(4)
NAME	NOT NULL	VARCHAR2(20)
DEPT	NOT NULL	VARCHAR2(5)
YEAR	NOT NULL	NUMBER(1)
HOSTEL		VARCHAR2(3)

SQL> select * from stu_details;

ROLLNO	NAME	DEPT	YEAR	HOS
-----	-----	-----	-----	-----
180	Yokarajan	IT	2	Yes
170	viji	CSE	3	Yes
137	Ros	AIDS	1	Yes
131	rithi	MTS	2	Yes
134	gojo	FT	3	Yes

DROP ATTRIBUTE:

SQL> alter table stu_details drop column hostel;
Table altered.

SQL> desc stu_details;

Name	Null?	Type
-----	-----	-----
ROLLNO	NOT NULL	NUMBER(4)
NAME	NOT NULL	VARCHAR2(20)
DEPT	NOT NULL	VARCHAR2(5)
YEAR	NOT NULL	NUMBER(1)

SQL> select * from stu_details;

ROLLNO	NAME	DEPT	YEAR
-----	-----	-----	-----
180	Yokarajan	IT	2
170	viji	cse	3
137	Ros	AIDS	1
131	rithi	MTS	2
134	gojo	FT	3

RENAMING THE ATTRIBUTE NAME:

SQL> alter table stu_details rename column name to stud_name;

Table altered.

SQL> select * from stu_details;

ROLLNO	STUD_NAME	DEPT	YEAR
-----	-----	-----	-----
180	Yokarajan	IT	2
170	viji	CSE	3
137	Ros	AIDS	1
131	rithi	MTS	2
134	gojo	FT	3

5 rows selected.

RENAMING THE TABLE :

SQL> alter table stu_details rename to student;

Table altered.

SQL> select * from student;

ROLLNO	STUD_NAME	DEPT	YEAR
180	Yokarajan	IT	2
170	viji	cse	3
137	Ros	AIDS	1
131	rithi	MTS	2
134	gojo	FT	3

5 rows selected.

3. TRUNCATING TABLES:

```
SQL> truncate table stud;
```

Table truncated.

```
SQL> select * from stud;
```

no rows selected

4. DROP:

```
SQL> drop table stud;
```

Table dropped.

AFTER DROPPING TABLE:

```
SQL> select * from stud;
```

```
select * from stud
```

*

ERROR at line 1:

ORA-00942: table or view does not exist

CONSTRAINTS:

1. PRIMARY KEY:

```
SQL> create table stud(id number(4) primary key, name varchar2(20) not null, cgpa
number(3,1) not null, dept varchar2(20) not null, address varchar2(20) unique, age
number(2) check (age >= 18));
```

```
SQL> insert into stud values(180, 'Yoks', 7.85, 'IT', 'Salem', 18);
```

```
insert into stud values(180, 'Yoks', 7.85, 'IT', 'Salem', 18)
```

*

ERROR at line 1:

ORA-00001: unique constraint (SYSTEM.SYS_C004092) violated

2.CHECK:

```
SQL> create table stud(id number(4) primary key, name varchar2(20) not null, cgpa
number(3,1) not null, dept varchar2(20) not null, address varchar2(20) unique, age
number(2) check (age >= 18));
```

Table created.

```
SQL> insert into stud values(111, 'Yoks', 7.85,'IT','Salem',17);
```

```
insert into stud values(111, 'Yoks', 7.85,'IT','Salem',17)
```

*

ERROR at line 1:

ORA-02290: check constraint (SYSTEM.SYS_C004091) violated

3.NOT NULL:

```
SQL> create table stud(id number(4) primary key, name varchar2(20) not null, cgpa
number(3,1) not null, dept varchar2(20) not null, address varchar2(20) unique, age
number(2) check (age >= 18));
```

Table created.

```
SQL> insert into stud values(111, 'Yoks', 7.85,'','Salem',18);
```

```
insert into stud values(111, 'Yoks', 7.85,'','Salem',18)
```

*

ERROR at line 1:

ORA-01400: cannot insert NULL into ("SYSTEM"."STUD"."DEPT")

4.UNIQUE:

```
SQL> create table stud(id number(4) primary key, name varchar2(20) not null, cgpa
number(3,1) not null, dept varchar2(20) not null, address varchar2(20) unique, age
number(2) check (age >= 18));
```

Table created.

```
SQL> insert into stud values(111, 'Yoks', 7.85,'IT','Karur',18);
```

```
insert into stud values(111, 'Yoks', 7.85,'IT','Karur',18)
```

*

ERROR at line 1:

ORA-00001: unique constraint (SYSTEM.SYS_C004093) violated

VALID STATEMENT:

```
SQL> select * from stud;
```

ID	NAME	CGPA	DEPT	ADDRESS	AGE
180	Yokarajan	7.9	IT	Karur	19
170	viji	8.5	cse	Chennai	18
137	Ros	7.8	AIDS	Erode	19

```
SQL> insert into stud values(111, 'Yoks', 7.85, 'IT', 'CBE', 18);
```

```
1 row created.
```

5.PRIMARY KEY AND FOREIGN KEY:

PARENT TABLE:STUDENT TABLE:

```
SQL> create table student(rollno number(4) primary key, name varchar2(20) not null, dept varchar2(20) not null, year number(1) not null);
```

CHILD TABLE:ACADEMICS TABLE:

```
SQL> create table academics(rollno number(4), cgpa number(3,1) not null, address varchar2(100), foreign key (rollno) references student(rollno));
```

INSERTING THE RECORD INTO CHILD TABLE :

ID EXISTS IN PARENT TABLE:

```
SQL> insert into academics values(&rollno, &cgpa, '&address');
```

```
Enter value for rollno: 180
```

```
Enter value for cgpa: 7.8
```

```
Enter value for address: Karur
```

```
old 1: insert into academics values(&rollno, &cgpa, '&address')
```

```
new 1: insert into academics values(180, 7.8, 'Karur')
```

```
1 row created.
```

ID NOT EXSITS IN PARENT TABLE:

```
Enter value for rollno: 218
```

```
Enter value for cgpa: 8.9
```

```
Enter value for address: Karur
```

```
old 1: insert into academics values(&rollno, &cgpa, '&address')
```

```
new 1: insert into academics values(218, 8.9, 'Karur')
```

```
insert into academics values(218, 8.9, 'Karur')
```

*

ERROR at line 1:

ORA-02291: integrity constraint (SYSTEM.SYS_C004095) violated - parent key not Found

REMOVE RECORD FROM PARENT TABLE:

```
SQL> delete from student where rollno=134;
```

```
delete from student where rollno=134
```

*

ERROR at line 1:

ORA-02292: integrity constraint (SYSTEM.SYS_C004095) violated - child record Found

DELETION USING ON DELETE CASCADE:

PARENT TABLE(PROJECT):

```
SQL> create table project(project varchar2(20), cname varchar2(20), budget number(12,2) check (budget > 0), primary key(project));
```

CHILD TABLE(EMPLOYEE):

```
SQL> create table employee(eid number(4), ename varchar2(20), project varchar2(20), salary number(8,2), primary key(eid), foreign key (project) references project(project) on delete cascade);
```

```
SQL> desc employee;
```

Name	Null?	Type
-----	-----	-----
EID	NOT NULL	NUMBER(4)
ENAME		VARCHAR2(20)
PROJECT		VARCHAR2(20)
SALARY		NUMBER(8,2)

INSERTING VALUES INTO CHILD TABLE:

```
SQL> insert into employee values(&eid, '&ename', '&project', &salary);
```

Enter value for eid: 101

Enter value for ename: YOKS

Enter value for project: Proj1

Enter value for salary: 40000

old 1: insert into employee values(&eid, '&ename', '&project', &salary)

new 1: insert into employee values(101, 'YOKS', 'Proj1', 40000)

1 row created.

AFTER INSERTING VALUES INTO CHILD TABLE:

SQL> select * from employee;

EID	ENAME	PROJECT	SALARY
101	YOKS	Proj1	40000
102	Bob	Proj2	62000
103	Charlie	Proj3	70000

DELETING THE RECORD IN THE PARENT TABLE:

SQL> delete from project where project='Proj3';

1 row deleted.

SQL> SQL> select * from project;

PROJECT	CNAME	BUDGET
Proj1	ClientA	50000
Proj2	ClientB	75000

DELETED RECORD IN CHILD TABLE:

SQL> select * from employee;

EID	ENAME	PROJECT	SALARY
101	YOKS	Proj1	40000
102	Bob	Proj2	62000

ON DELETE SET NULL:

SQL> create table employee(eid number(4), ename varchar2(20), project varchar2(20), salary number(8,2), primary key(eid), foreign key (project) references project(project) on delete set null);

Table created.

INSTANCE OF CHILD TABLE:

```
SQL>Select * from employee;
```

EID	ENAME	PROJECT	SALARY
101	Alice	Proj1	55000
102	Bob	Proj2	62000

PARENT TABLE:

```
SQL> Select * from project;
```

PROJECT	CNAME	BUDGET
Proj1	ClientA	50000
Proj2	ClientB	75000

DELETING VALUE IN PARENT TABLE:

```
SQL> delete from project where project='Proj2';
```

1 row deleted.

AFTER DELETING VALUES IN PARENT TABLE:

```
SQL> Select * from project;
```

PROJECT	CNAME	BUDGET
Proj1	ClientA	50000

AFTER DELETING VALUES IN CHILD TABLE:

```
SQL> Select * from employee;
```

EID	ENAME	PROJECT	SALARY
101	Alice	Proj1	55000
102	Bob		62000

ADD AND DROP CONSTARINTS:

ADDING CONSTRAINTS IN SCHEMA:

BEFORE ADDING CONSTRAINTS:

SQL> desc emp;

Name	Null?	Type

EID		NUMBER(3)
NAME	NOT NULL	VARCHAR2(20)
SALARY		NUMBER(15)

SQL> alter table emp add constraint pkid primary key(eid);

Table altered.

AFTER ADDING CONSTRAINTS:

SQL> desc emp;

Name	Null?	Type

EID	NOT NULL	NUMBER(3)
NAME	NOT NULL	VARCHAR2(20)
SALARY		NUMBER(15)

ADDING CONSTRAINTS IN INSTANCE:

SQL> alter table emp add constraint uni unique(salary);

alter table emp add constraint uni unique(salary)

*

ERROR at line 1:

ORA-02299: cannot validate (SYSTEM.UNI) - duplicate keys found

DROP CONSTRAINTS:

SQL> alter table emp drop constraint pkid;

Table altered.

SQL> desc emp;

Name	Null?	Type

EID		NUMBER(3)
NAME	NOT NULL	VARCHAR2(20)
SALARY		NUMBER(15)

CONTENTS	MARKS ALLOTED	MARKS OBTAINED
Aim/Algorithm, Sqlpl/Sqlite	30	
Execution And Result	20	
Viva	10	
Total	60	

RESULT:

Thus Data Definition Language Commands and Integrity Constraints were executed.